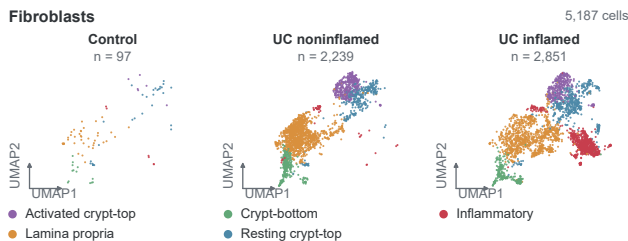
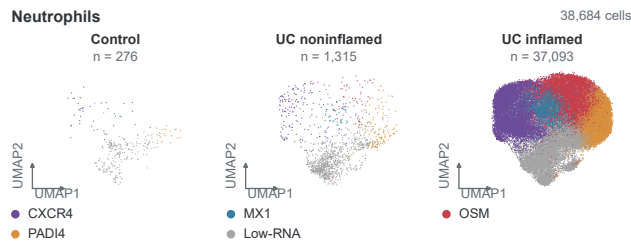


## A Human UC fibroblast and neutrophil state landscape

### Fibroblasts



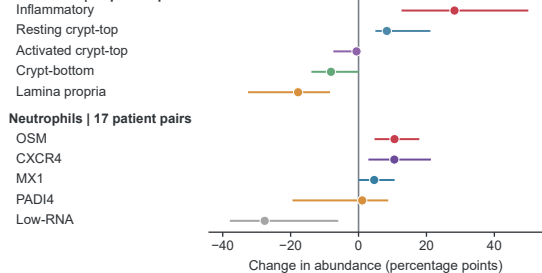
### Neutrophils



## B Paired cell-state remodeling

Inflamed minus noninflamed; median and 95% CI

### Fibroblasts | 13 patient pairs

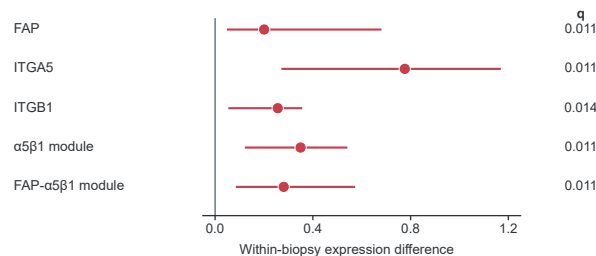


### Neutrophils | 17 patient pairs



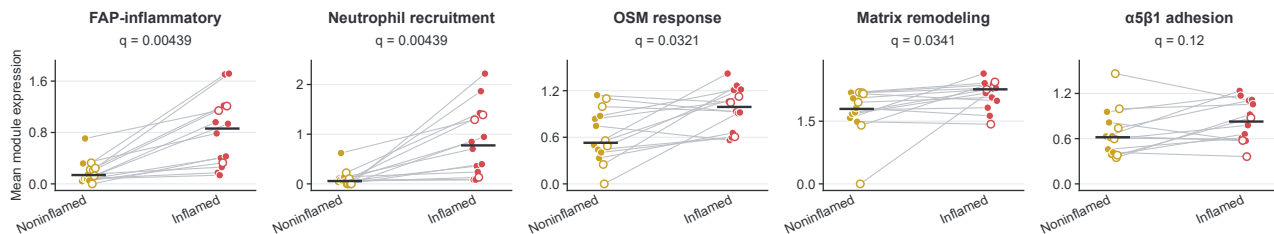
## C Inflammatory-fibroblast activation

Inflammatory vs other fibroblasts; 10 biopsies



## D Paired fibroblast program changes

14 patient pairs; lines connect patients; black bars, medians; open points, <10 recovered cells



## E Neutrophil state-associated RNA programs

11 matched patients; ≥20 cells/state; naming genes excluded



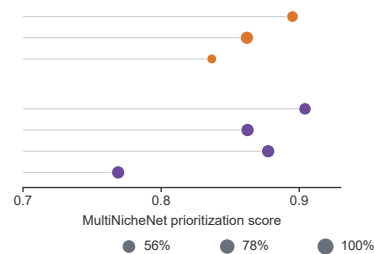
## F Predicted reciprocal signals

### Fibroblast → neutrophil

HGF → CD44  
CXCL1 → CXCR2  
CSF3 → CSF3R

### Neutrophil → fibroblast

OSM → OSMR  
IL1B → IL1R1  
NAMPT → ITGA5  
NAMPT → ITGB1



LR-expressing fraction

56% 78% 100%

Equal patient weights; Friedman q (BH, 9 programs).

Paired coexpression: Supplementary Figure 1C.